1 Visualizing multivariable functions

1.1 Shapes and functions

Consider the following three expressions:

$$(\cos(t), \sin(t))$$
$$x^2 + y^2 = 1$$
$$\sqrt{1 - x^2}$$

Each evokes a shape: a circle. (The last may evoke only an arc of a circle.) Let us describe each of these evocations in the language of sets and functions: for each, we define sets X, Y, a function $f : X \to Y$, and explain how the shape appears as a subset of either Y, X, or $X \times Y$.

- *Image*. As we have seen before, take $X = [0, 2\pi)$, $Y = \mathbb{R}^2$, and define

$$f : [0, 2\pi) \to \mathbb{R}^2$$
$$t \mapsto (\cos(t), \sin(t)).$$

  Then the circle is the *image* $f([0, 2\pi)) \subseteq Y$ of the function f, a subset of the codomain. Note that the image construction parametrizes a shape.

- *Preimage*. Now take $X = \mathbb{R}^2$, $Y = \mathbb{R}$ and define

$$f : \mathbb{R}^2 \to \mathbb{R},$$
$$(x, y) \mapsto x^2 + y^2.$$

  Then the circle is the *preimage* $f^{-1}(1) \subseteq X$, a subset of the domain.

- *Graph*. Take $X = [-1, 1]$ and $Y = \mathbb{R}$. Define

$$f : [-1, 1] \to \mathbb{R}$$
$$x \mapsto \sqrt{1 - x^2}.$$

  Then a half-circle is the graph $\Gamma(f) \subseteq X \times Y$, the subset of the Cartesian product defined by

$$\Gamma(f) = \{(x, f(x)) : x \in X\}.$$

In the following, we will use each of these constructions to visualize a multivariable function.

1.2 Graphs

Definition 1.1 (Graph)**.** Given a function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$, we define the graph in essentially the same way as in one variable calculus:

$$\Gamma(f) := \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in D\} \subseteq \mathbb{R}^n \times \mathbb{R} \cong \mathbb{R}^{n+1}.$$

When $n = 2$, $\Gamma(f) \subseteq \mathbb{R}^3$, which we can sketch as a surface in three-dimensional space. Here is a useful fact:

Proposition 1.2. *The graph of a function $f(x,y)$ which only depends on x and y in the combination x^2+y^2 is rotationally symmetric about the z-axis.*

Another technique to help us visualize the graph of a multivariable function is to consider *cross-sections*.

Definition 1.3 (Cross-section)**.** A *cross-section* (or just *section*) of the graph of a function $z = f(x,y)$ is the intersection of the graph with a plane.

Example 1.4. Consider the function $f(x,y) = x^2+y^2$. The intersection of the graph of f with the xz-plane is the parabola $f(x,0) = x^2$. Since the graph of f is rotationally symmetric, the graph is obtained by rotating the parabola $z = x^2$ about the z-axis. The resulting graph is called a *paraboloid*.

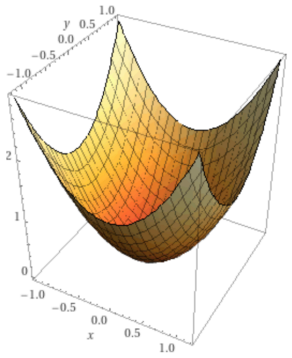


Figure 1: Graph of the paraboloid $f(x,y) = x^2+y^2$.

Exercise 1.5. Sketch the graph of the function $f(x,y) = 9-(x-1)^2-(y+3)^2$.

Take the graph of the paraboloid $f(x,y) = x^2+y^2$, reflect it across the xy-plane, translate it up 10 units, and translate the vertex to the point $(1,-3)$. The intersection of the graph with the plane parallel to the xz-plane passing through $(1,-3)$ is the parabola $f(x,y) = 9-(x-1)^2$. This parabola is concave down with vertex at $(x,z) = (-1,10)$. Rotating this parabola about the line through $(-1,10)$ parallel to the z-axis gives the graph of f. Note that it intersects the xy-plane along the circle $(x-1)^2+(y+3)^2 = 9$ of radius 3 centered at $(1,-3)$.

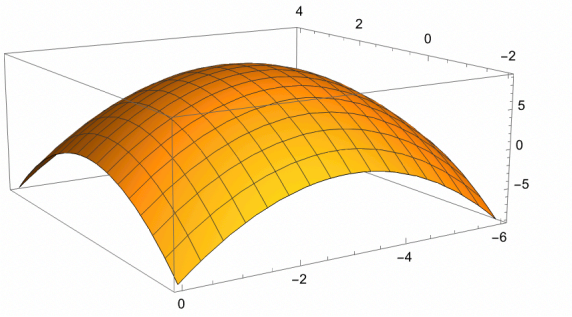


Figure 2: Graph of the paraboloid $f(x,y) = 9-(x-1)^2-(y+3)^2$.

Exercise 1.6. Sketch the graph of the distance function $\rho(x,y) = \sqrt{x^2+y^2}$.

Proof. The intersection with the xz-plane is the graph of $f(x,0) = \sqrt{x^2} = |x|$. Rotating the graph of $y = |x|$ about the z-axis gives the graph of the distance function. □

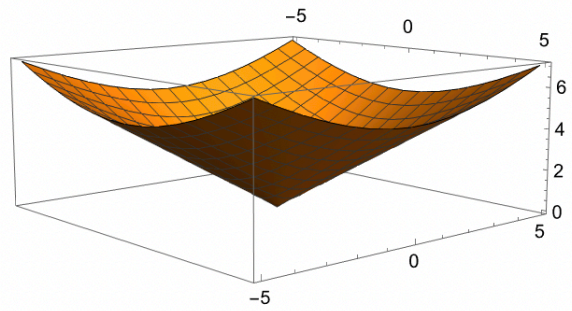


Figure 3: Graph of the distance function $\rho(x,y) = \sqrt{x^2+y^2}$.

1.3 Level curves

We now visualize the graph of a multivariable function using the preimage construction. We will consider the image construction (parametrized surfaces) when we study vector calculus in a later unit.

Definition 1.7 (Level curve, contour line)**.** A *level curve* of f is the set

$$L(f) = \{(x,y) \in U \subset \mathbb{R}^2 : f(x,y) = \text{constant}\}$$

To each level curve in the domain we associate a *contour line*, which is the intersection of the plane $z = c$ with the surface $z = f(x,y)$.

Example 1.8. The graph of the function $z = 100 - x^2 - y^2$ is shown below, along with the contour line at $z = 75$, and the corresponding level curve.

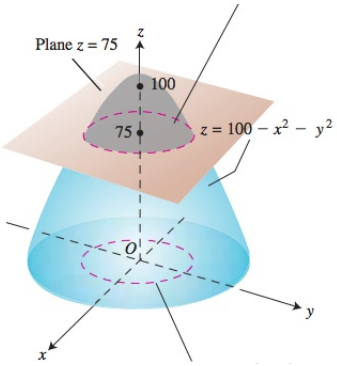
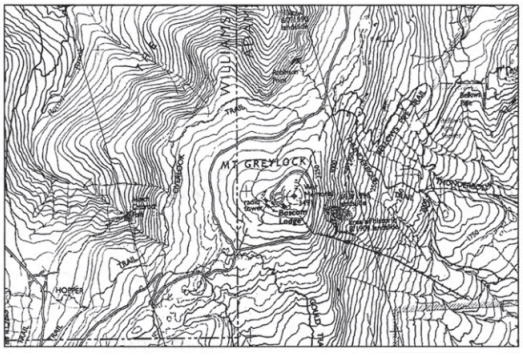


Figure 4: Graph of $z = 100 - x^2 - y^2$. The contour line at $z = 75$ and corresponding level curve are shown.

Example 1.9 (Elevation maps)**.** On maps, contour lines (really level curves; many people use these terms interchangibly) represent constant elevation.



Example 1.10. The level curves of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) = x^2 + y^2$ are concentric circles centered at the origin.

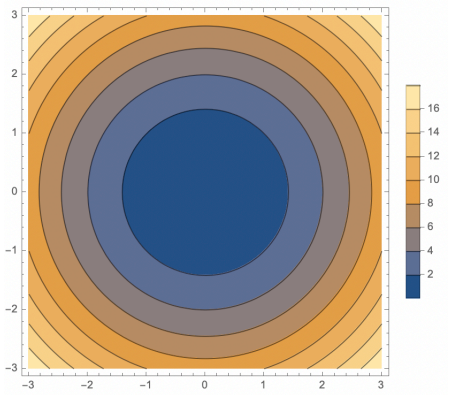


Figure 5: The level curves of the paraboloid $z = x^2 + y^2$. The shading indicates the elevation of the corresponding contour lines.

More generally, for functions $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ with $n > 2$, one defines the *level set at height c* as $f^{-1}(c)$, even though we cannot visualize these.

Example 1.11. The *level surfaces* of the function $f : \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x,y,z) = x^2 + y^2 + z^2$ are concentric spheres, centered at the origin.

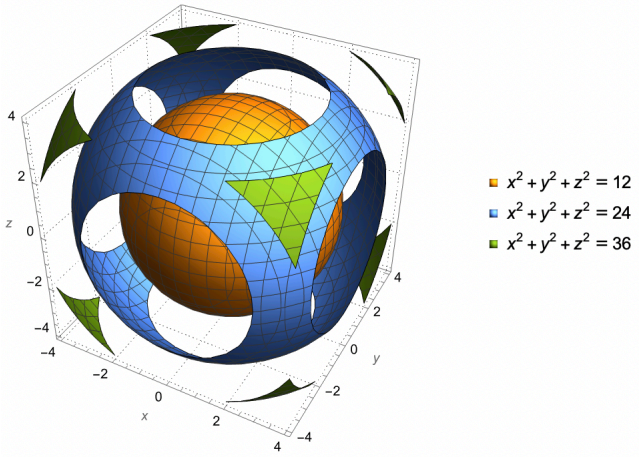


Figure 6: The level surfaces of the function $f(x,y,z) = x^2 + y^2 + z^2$.

Exercise 1.12. Describe the level surfaces of the function $F : \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x,y,z) = x + y + z$.

Given $c \in \mathbb{R}$, $f^{-1}(c) = \{(x,y,z) : x + y + z = c\}$, which is a plane parallel to the vector $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$. Thus, level sets form a family of parallel planes with normal vector $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$.